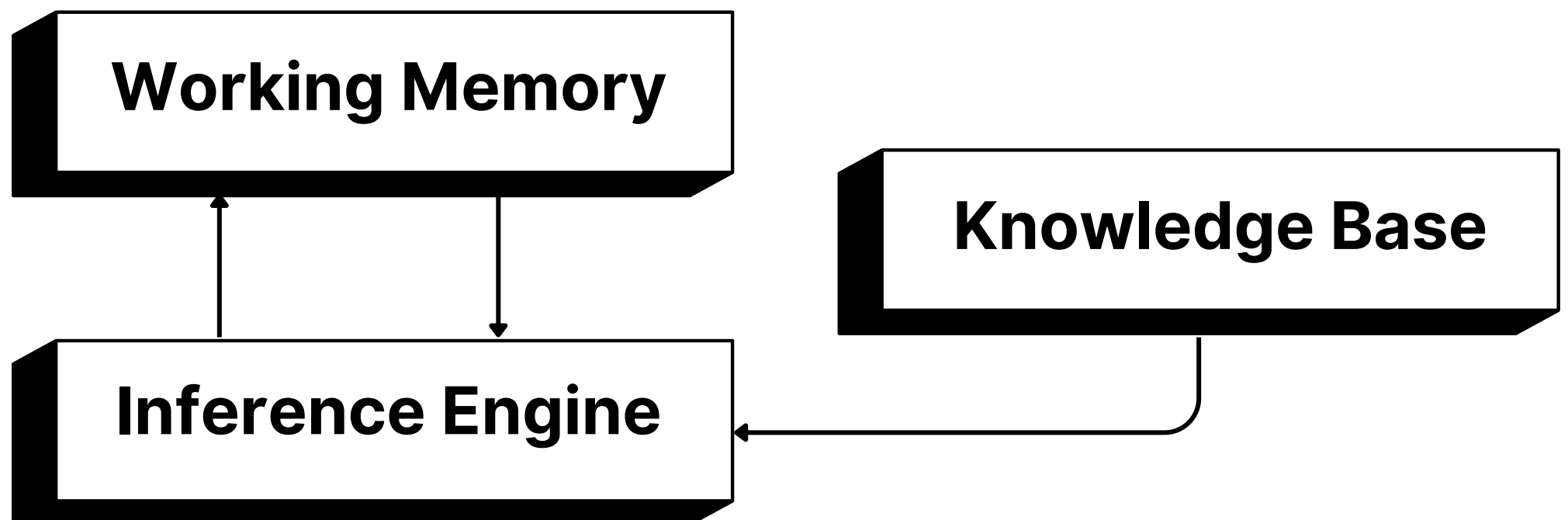
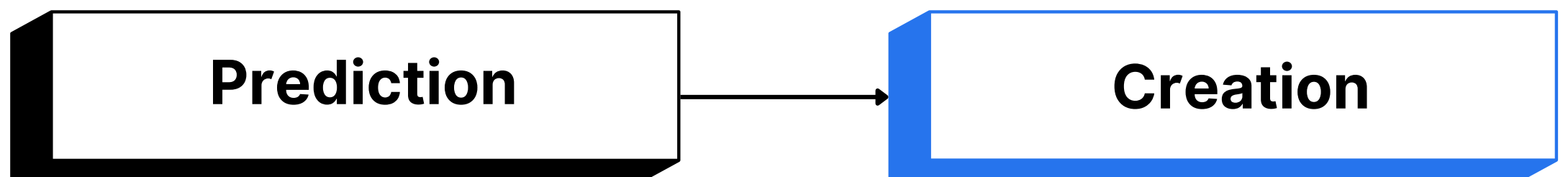


Evolution of AI

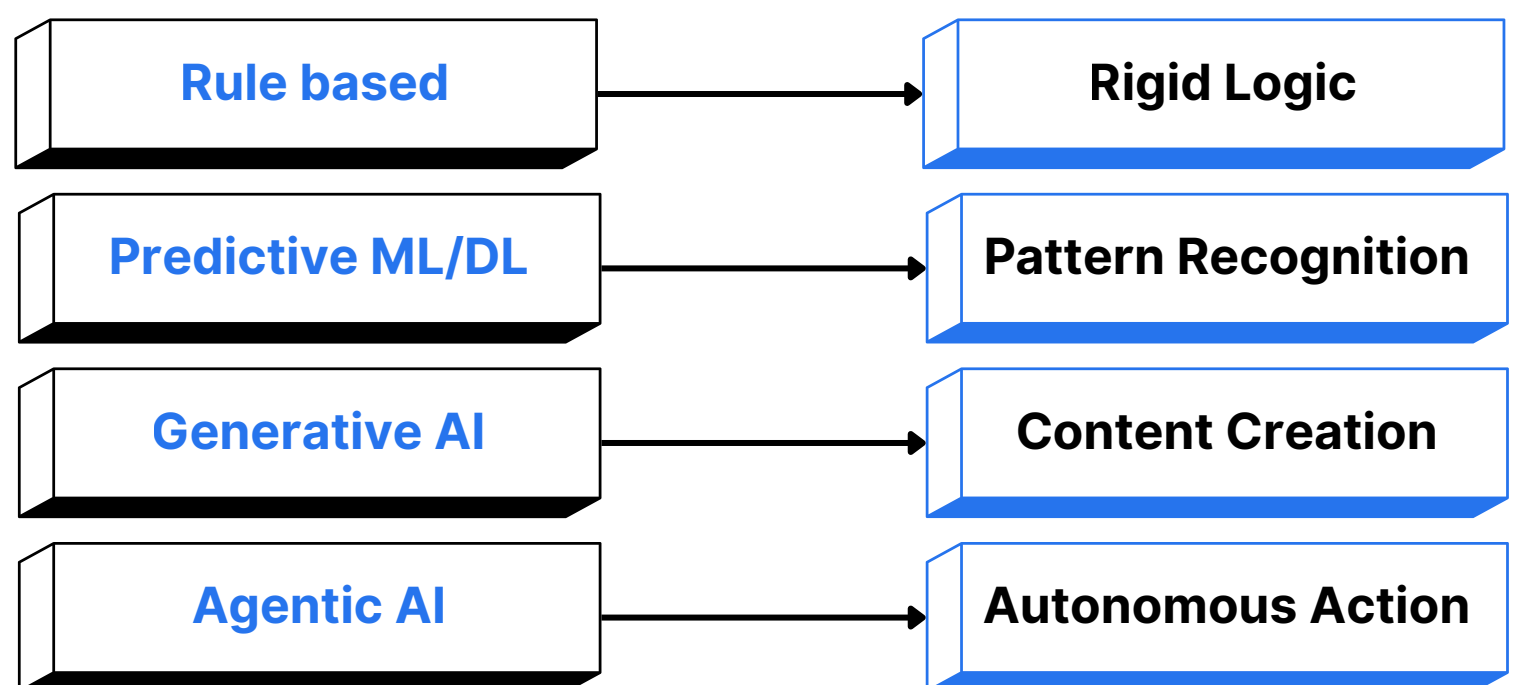
Rule-based Systems



Generative AI



BIG
Picture



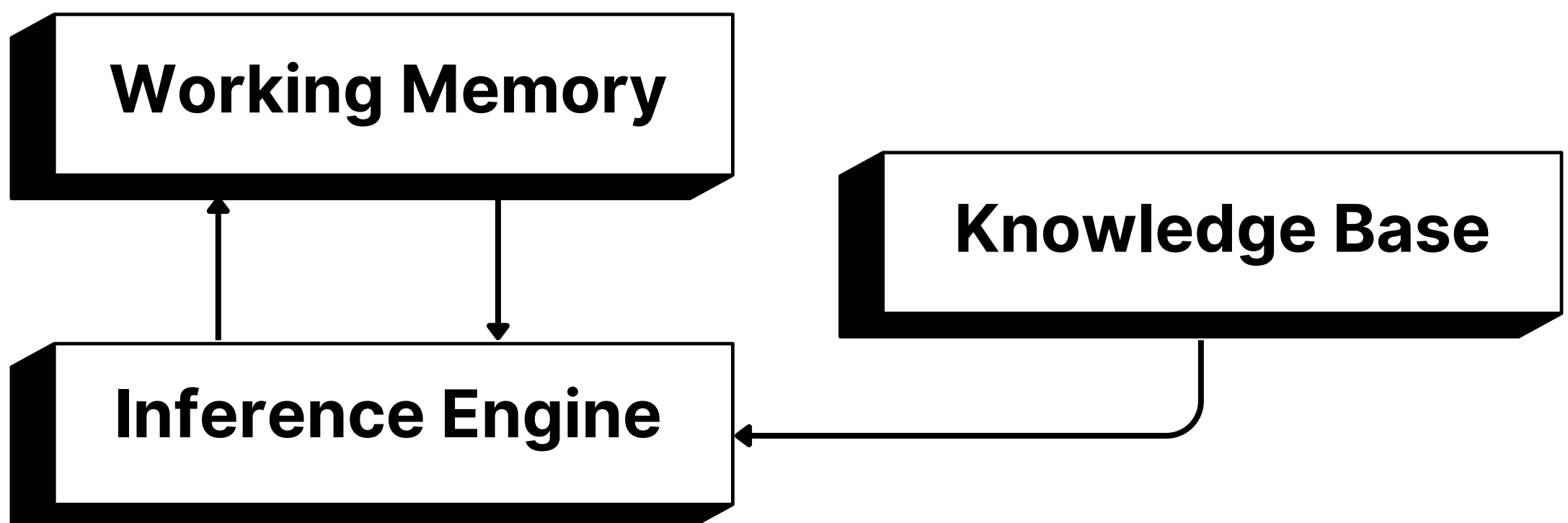
#30DaysofAgenticAI

Rule-based Systems

The earliest AI systems were built on strict rules and logic. Think of expert systems in the 1980s: they relied on human-coded “if-then” instructions to simulate decision-making. They were powerful in narrow, well-defined domains but brittle outside them. Any situation not explicitly covered by rules broke the system.

Mental model: Symbolic.

Knowledge was represented as symbols and manipulated by logic. Machines were essentially executing structured reasoning.



Predictive ML/DL

With more data and compute, machine learning and deep learning shifted AI away from hand-coded rules. Instead of hardwiring knowledge, models learned patterns from data. This unlocked progress in image recognition, speech processing, and recommendation engines.

Mental model: Neural.

Instead of explicitly representing symbols, systems learned statistical associations in a high-dimensional space.

What changed was adaptability. Rule-based systems needed engineers to write new logic. Predictive models adapted by training on new data. Still, these systems were narrow: they solved specific problems like “classify this image” or “predict the next word.”



Generative AI

Generative models like GANs, GPT and Diffusion based systems moved AI from prediction to creation. They don't just classify or predict, they generate text, images, music, and more. They can generalize across tasks better than predictive models, thanks to training on massive, diverse datasets.

Mental model: Emergent/Foundational model.

The outputs feel symbolic (language, images), but the engine is entirely neural. The gap between symbolic reasoning and neural generalization is starting to close.

The leap here was from automation to creation. AI stopped being just a tool for prediction and became a partner in producing new content.



Agentic AI

Now we're entering the agentic stage. Agentic AI doesn't just generate it. These systems can plan, reason over multiple steps, call tools, and pursue goals autonomously. Instead of a passive system waiting for prompts, agentic AI can initiate, coordinate, and adapt in real-world workflows.

Mental model: Hybrid.

Symbolic-like planning is layered on top of neural foundations. The system isn't just an autocomplete engine, it's closer to an autonomous problem-solver.

The shift is from narrow automation to autonomous, goal-oriented systems. Rule-based systems automated tasks step by step. Predictive ML automated recognition. Generative AI automated creation. Agentic AI is automating the agency itself: deciding what needs to be done and taking action toward it.



Big Picture

The trajectory is clear:

- Rule-based: rigid logic
- Predictive ML/DL: pattern recognition
- Generative AI: content creation
- Agentic AI: autonomous action

Symbolic vs neural has been the core mental model divide, and agentic AI is starting to merge the strengths of both. What this really means is we're moving from tools that *assist* to systems that can *collaborate and operate independently*.